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Multi-source data integration through pooling and transfer learning improves generalizability and specialization of deep learning models for ICU mortality and length of stay prediction: a four-database external validation study

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Abstract

Background Most Artificial Intelligence prognostic models in intensive care are trained and validated on a single source, limiting their reliability in external settings. This study evaluated generalizability and specialization for ICU mortality and remaining length of stay (RLoS) prediction across international databases, and explored multi-source strategies to mitigate the performance cost of external deployment.

Methods Using the Temporal Pointwise Convolution (TPC) architecture, we conducted external validation across four harmonized BlendedICU databases (eICU-CRD, MIMIC-IV, AmsterdamUMCdb, HiRID), extracting nearly 20,000 unique patients from each. Four training configurations were compared: single-source training on AmsterdamUMC ($N=3,574$) or MIMIC-IV ($N=10,915$), data pooling, and transfer learning (both $N=14,489$). Performance was assessed using AUROC and AUPRC for mortality, MAPE for RLoS, and a composite metric ($M_{composite}$).

Results Internal validation consistently overestimated external performance, with MAPE increasing by up to 51.8% and AUROC dropping by up to 13.8% on external databases. Site-specific features such as drug exposure acted as “shortcuts” degrading portability. Data pooling emerged as the superior strategy, achieving the best generalization ($M_{composite}$ improvement up to 8.0% over single-source models) while matching or outperforming transfer learning for specialization — particularly for RLoS (MAPE 84.49 vs. 89.24, non-overlapping confidence intervals).

Conclusion Single-source training compromises clinical applicability. Data pooling across harmonized international cohorts is the most effective strategy for both generalization and specialization, supporting a “Data-Centric AI” approach. We advocate for the democratization of local ICU datasets and open-source customizable models to empower independent clinical validation.

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Keywords Intensive care unit, Length of stay, Artificial intelligence, Deep learning, Machine learning, Prediction, Validation, Bias

Introduction

Predictive models based on machine learning have shown considerable promise in intensive care unit (ICU) applications, ranging from mortality risk assessment to treatment response classification [1–3]. However, a recurring weakness in the literature is the reliance on single-center datasets for both model derivation and validation [4–6]. This methodological choice is not merely a matter of convenience; it reflects a fundamental challenge in critical care data science: the lack of harmonization among major open-access databases. This fragmentation severely limits the feasibility of multi-center validation and hinders the assessment of cross-dataset generalization. While internal validation techniques are common, they often demonstrate a model's ability to fit retrospective data rather than its utility in clinical practice, where data distributions may significantly deviate from the training set [7, 8]. This lack of proven transferability remains a major barrier to the clinical deployment of Artificial Intelligence.

The proliferation of electronic health records has led to several high-quality ICU research databases, such as eICU and MIMIC-IV from the United States, AmsterdamUMC from the Netherlands, and HiRID from Switzerland [9–12]. Despite the massive volume of available data, these sources remain largely siloed. The BlendedICU dataset addresses this gap by providing harmonized data across these four major repositories, enabling robust multi-source training and external validation strategies [13]. Despite these resources, external validation has historically been scarce; estimated at only 9% for mortality models as of 2021, and virtually non-existent for Remaining Length of Stay (RLoS) [14]. While recent studies have begun to address this gap by conducting cross-database validation across the same four international ICU databases for a range of clinical outcomes including sepsis, acute kidney injury, and length of stay, a systematic and quantified characterization of the performance cost incurred during external validation; and of the strategies best suited to mitigate it; remains lacking [6, 15–17]. Consequently, clinicians lack sufficient data on the model's performance across different healthcare systems, where variations in equipment, clinical protocols, and patient demographics; known as “*domain shift*”; can significantly reduce the accuracy of predictions [5, 18, 19]. Beyond a statistical concern, when a model is deployed at a new institution without external validation, its predictions may reflect spurious associations specific to the training site - such as local prescribing habits, measurement frequencies, or case-mix characteristics - rather than generalizable physiological signals. When clinicians

act on such predictions to adjust monitoring intensity, escalate or de-escalate treatment, or allocate scarce ICU beds, errors arising from undetected domain shift can lead to both over- and under-treatment, with potentially life-threatening consequences in a setting where decisions must be made rapidly and under uncertainty.

To address these critical gaps, we present a comprehensive investigation of multi-source data integration to enhance both the generalization (performance on unseen institutions) and specialization (performance on a target institution) of deep learning models. Specifically, we focus on two clinically vital outcomes: ICU mortality and RLoS. Unlike static admission-time prediction scores, our model updates predictions continuously and hourly throughout the ICU stay, providing a dynamic trajectory of risk rather than a single estimate of a non-actionable outcome [20]. In this context, a rising mortality probability can prompt escalation of monitoring or goals-of-care discussions; a decreasing remaining length of stay can trigger early discharge planning and bed management. From the patient and family perspective, anticipated duration of ICU stay is among the most frequently asked questions at admission. From an institutional standpoint, real-time length-of-stay forecasting is a key lever for bed management - a role that was made starkly evident during the COVID-19 pandemic, when predicted ICU occupancy directly informed capacity planning decisions across health systems.

Leveraging the harmonized BlendedICU framework and a deep learning model trained on hourly-sampled tabular time series data, this study aims to: (1) evaluate the generalizability of RLoS and mortality predictions by conducting the first systematic quantification of the performance cost; (2) compare multi-source training strategies, specifically analyzing the trade-offs between single-dataset training, data pooling, and transfer learning in terms of predictive performances; and (3) assess the impact of data volume by varying the size of training samples to determine the amount of data required (cost/benefit balance).

By addressing these objectives, this work seeks to provide practical insights into the requirements for deploying AI models that remain reliable across diverse institutional contexts, moving beyond retrospective performance toward clinical applicability.

Methods

This study is reported according to the TRIPOD + AI guidelines [21]. All code is available at <https://github.com/CIC-1410/BlendedLoS>.

Ethics statement

Datasets were observational and had preexisting ethical approvals or exemptions that allowed for retrospective analysis [9–12]. Access was obtained following the respective requirements for each dataset.

Data source

We used the BlendedICU dataset, which harmonizes data from four major publicly available ICU databases: MIMIC-IV (USA), eICU-CRD (USA), HiRID (Switzerland), and AmsterdamUMCdb (Netherlands) [9–13]. Nearly 20,000 unique patients were extracted from each of the four subsets, yielding a total of approximately 80,000 patients. Input variables included hourly-sampled vital signs, laboratory values, drug exposures, and clinical scores derived from electronic health records (as detailed in Additional File 1). Diagnoses were not included, as they are not consistently available across all four databases.

Data preprocessing

Missing data were handled using forward-filling combined with missingness indicator masks, a strategy appropriate for the Missing Not At Random (MNAR) structure of ICU data [22–24]. Prior to normalization, extreme values were clipped to physiologically plausible ranges, limiting the influence of outliers on model training. Full details of the preprocessing pipeline, including distribution statistics before and after clipping, are provided in Supplementary Table S1 (Additional File 1).

Outcome definition

Two outcomes were predicted jointly at hourly intervals throughout each ICU stay: in-ICU mortality (defined as death before ICU discharge) and remaining length of stay (RLoS), defined as time to discharge conditional on survival. Outcomes were predicted from 5 h after admission onward.

Model

We used the Temporal Pointwise Convolution (TPC) architecture, a deep learning model specifically designed for ICU time series data that has demonstrated superior performance over alternative architectures for joint mortality and RLoS prediction [25]. Prior to adopting TPC, we benchmarked three state-of-the-art architectures - Transformer, Long Short-Term Memory (LSTM), and TPC - on our two-stage prediction task across all four BlendedICU databases. TPC outperformed both alternatives on most subsets and both prediction tasks; full results are provided in Additional File 2 (Table S2). Data were split at the patient level into training (70%), validation (15%), and test (15%) sets to prevent data leakage.

Performance evaluation

Model discrimination for mortality was assessed using the Area Under the Receiver Operating Characteristic curve (AUROC) and the Area Under the Precision-Recall Curve (AUPRC). RLoS accuracy was assessed using the Mean Absolute Percentage Error (MAPE). A composite metric (*Mcomposite*) was introduced to jointly summarize performance across both tasks, normalized to the internal validation baseline of each model: a value below 1 indicates improvement over the internal baseline, and a value above 1 indicates degradation. The composite metric was used for cross-configuration ranking only; AUROC and MAPE remain the primary performance metrics.

Specialization and generalization

Two properties were evaluated. Specialization refers to model performance on a site seen during training - the most relevant property for deploying a model in a known clinical environment. Generalization refers to performance on entirely unseen sites - the key property for a model intended for broad deployment across institutions. Four training configurations were compared: single-source training on AmsterdamUMC alone ($N=3,574$), single-source training on MIMIC-IV alone ($N=10,915$), transfer learning (pre-training on MIMIC-IV then fine-tuning on AmsterdamUMC, $N=14,489$), and data pooling (joint training on both, $N=14,489$). Generalization was strictly evaluated on the held-out eICU and HiRID databases, which were never used during training. The full study pipeline is illustrated in Fig. 1. A detailed description of all methodological steps, including preprocessing, model architecture, loss function, and statistical methods, is provided in Additional File 1.

Results

Patient stay

Table 1 presents certain characteristics of the 19,573 patients included in the present study and their ICU stays according to their inclusion in the four open databases. Some characteristics varied considerably between the original databases, sometimes by a factor of two such as ICU mortality (5.5% in eICU and 9.8% in AmsterdamUMC), or length of stay in ICU (49.5 [standard deviation = 64.2] hours in HiRID and 77.9 [standard deviation = 99.0] hours in AmsterdamUMC). Similarly, we observed great variations in medication exposure, such as the use of dopamine, which ranged from 0% in HiRID to 31% at AmsterdamUMC, or norepinephrine, which ranged from 11.6% in eICU to 35.4% at AmsterdamUMC. Distribution summary statistics before and after clipping of extreme values to physiologically plausible ranges and normalization-space clipping are provided in Additional File 1, Table S1.

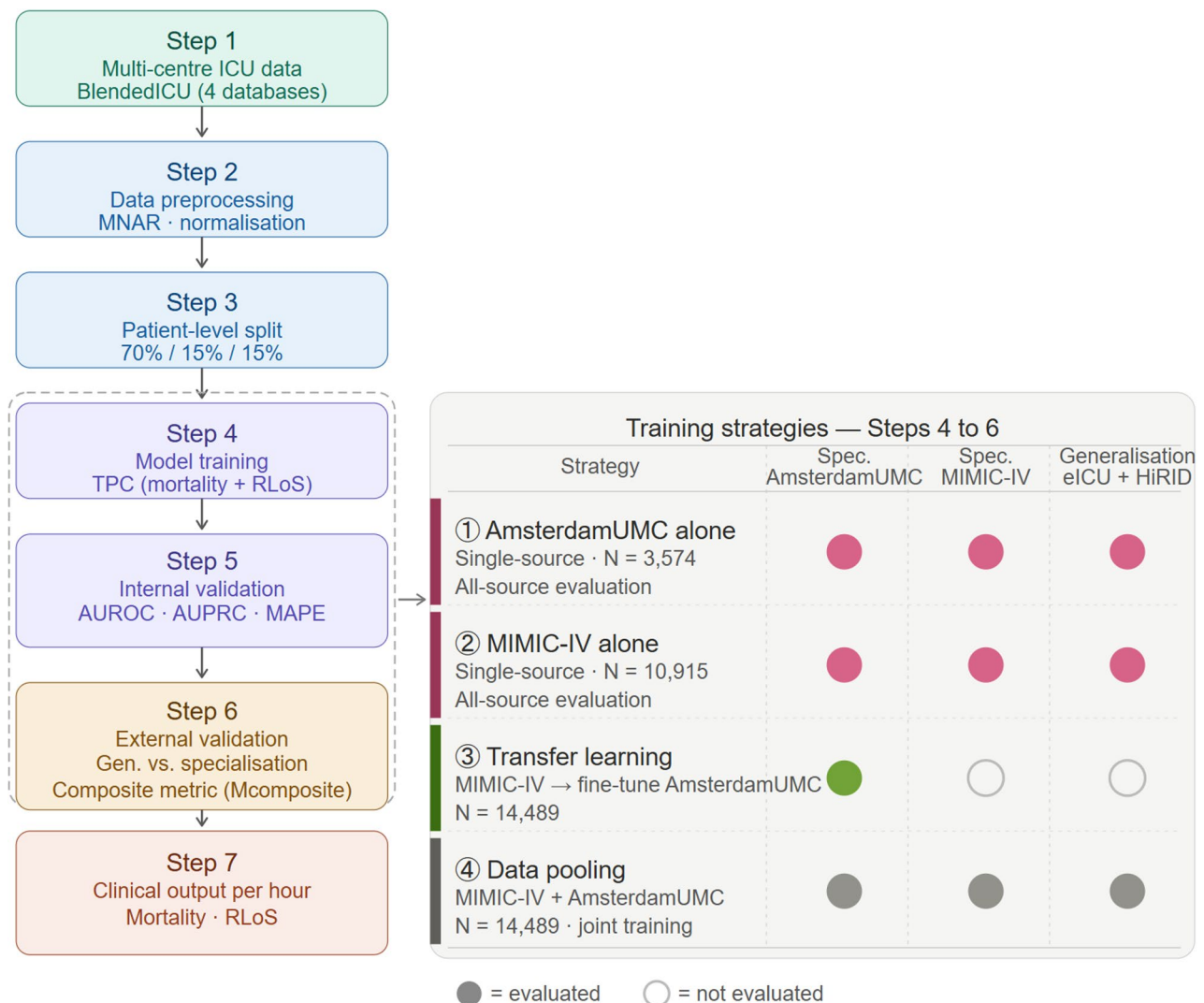


Fig. 1 Overview of the study pipeline and training strategies. The workflow proceeds in seven sequential steps (left panel). Step 1: data were extracted from the BlendedICU framework, which harmonizes four open-access international ICU databases (MIMIC-IV, eICU-CRD, HiRID, and AmsterdamUMCdb). Step 2: preprocessing included clipping of extreme values to physiologically plausible ranges, handling of Missing Not At Random (MNAR) data using forward-filling imputation with missingness indicator masks, and normalisation. Step 3: data were split at the patient level (training 70% / validation 15% / test 15%) to prevent data leakage. Step 4: a Temporal Pointwise Convolution (TPC) deep learning model was trained to jointly predict in-ICU mortality and remaining length of stay (RLoS). Step 5: internal validation was performed on the held-out test set from the same database, using the Area Under the Receiver Operating Characteristic curve (AUROC) and the Area Under the Precision-Recall Curve (AUPRC) for mortality, and the Mean Absolute Percentage Error (MAPE) for RLoS. Step 6: external validation was performed across four training strategies (right panel): (①) single-source training on AmsterdamUMC alone ($N = 3,574$); (②) single-source training on MIMIC-IV alone ($N = 10,915$); (③) transfer learning, with pre-training on MIMIC-IV followed by fine-tuning on AmsterdamUMC ($N = 14,489$) - evaluated for specialization on AmsterdamUMC only; (④) data pooling, with joint training on both MIMIC-IV and AmsterdamUMC ($N = 14,489$) - evaluated on all three sets. Specialization refers to performance on a site seen during training; generalization refers to performance on the unseen eICU and HiRID databases. A composite metric ($M_{composite}$) was introduced to jointly summarize mortality and RLoS performance relative to the internal validation baseline. Step 7: the model produces two outputs, updated hourly for each patient - predicted in-ICU mortality risk (probability) and predicted remaining length of stay (days, conditional on survival). AUROC: Area Under the Receiver Operating Characteristic curve; AUPRC: Area Under the Precision-Recall Curve; MAPE: Mean Absolute Percentage Error; MNAR: Missing Not At Random; RLoS: Remaining Length of Stay; TPC: Temporal Pointwise Convolution

Comparison of the internal and external validations

A comparison of the internal and external validations of the TPC model with training on each data source is shown in Table 2. Internal validation generally overestimated the model's capability on external data, although one exception was observed: the model trained on

AmsterdamUMC showed a +4.4% improvement in AUROC on external data, suggesting that this training source may generalize particularly well to other databases. The MAPE of RLoS prediction varied from -0.3% to $+51.8\%$ depending on the training set used; AUROC dropped by up to 13.8% for mortality prediction, with

Table 1 Characteristics of ICU patient stays included in the study, according to the original database

Characteristics	eICU	AmsterdamUMC	HiRID	MIMIC-IV
Multi-centric database	Yes (N = 238)	No	No	No
Patients, n	19,573	19,573	19,573	19,573
ICU Stays, n ¹	27,515	20,751	19,573	28,227
Demographics*				
Age (years), mean [SD]	63.2 [17.4]	56.5 [17.4]	63.6 [15.2]	63.4 [17.2]
Sex, n (%)				
• Female	8,887 (45.4%)	6,667 (34.1%)	6,955 (35.5%)	8,664 (44.3%)
• Male	10,672 (54.5%)	12,456 (63.6%)	12,618 (64.5%)	10,909 (55.7%)
• Unknown	14 (0.1%)	450 (2.3%)	-	-
Height (cm), mean [SD]	169.4 [13.5]	170.6 [8.7]	171.2 [8.6]	168.9 [15.6]
Weight (kg), mean [SD]	83.6 [26.0]	76.0 [13.5]	76.8 [15.2]	81.6 [49.0]
ICU LoS (hours), mean [SD]	61.6 [68.9]	77.9 [99.0]	49.5 [64.2]	75.6 [78.5]
ICU LoS > 10 days, n (%)	1,192 (4.3%)	2,387 (11.5%)	674 (3.4%)	1,851 (6.6%)
ICU mortality, n (%)	1,521 (5.5%)	2,033 (9.8%)	1,209 (6.2%)	2,061 (7.3%)
Continuous Variables				
Vitals, mean [SD]				
• Heart rate (/min)	87.0 [19.3]	86.0 [19.8]	84.6 [17.2]	87.1 [19.5]
• Invasive Systolic Arterial pressure (mmHg)	117.7 [20.1]	129.9 [28.5]	119.7 [24.9]	121.1 [24.4]
• O ₂ Saturation in Arterial blood (%)	96.3 [2.9]	91.9 [10.0]	96.5 [3.06]	82.3 [16.8]
• Temperature (°C)	37.1 [0.9]	36.6 [0.9]	37.19 [0.9]	37.0 [0.7]
Laboratory, mean [SD]				
• pH	7.37 [0.1]	7.38 [0.09]	7.42 [0.07]	7.38 [0.09]
• Platelets count (G/L)	198.2 [19.3]	198.1 [128.8]	190.2 [126.1]	195.6 [119.2]
Unit Type ² n (%)				
• Cardiac	3,847 (13.98%)	1,222 (5.89%)	-	7,619 (26.99%)
• Gastrosurgery	-	1,200 (5.78%)	-	-
• Medical	2,350 (8.54%)	1,013 (4.89%)	-	6,020 (21.33%)
• Medical-Surgical	15,521 (56.41%)	-	-	4,954 (17.55%)
• Neurological	1,996 (7.25%)	2,889 (13.92%)	-	1,885 (6.68%)
• Other	-	3,937 (18.97%)	-	-
• Surgical	3,801 (13.81%)	8,234 (39.68%)	-	4,348 (15.4%)
• Trauma	-	1,095 (5.28%)	-	3,401 (12.05%)
• Unknown	-	1,013 (5.59%)	19,573 (100%)	-
Drug Exposure ³ , n (%)				
• Dobutamine	394 (1.4%)	1,283 (6.2%)	2,542 (13.0%)	394 (1.4%)
• Dopamine	601 (2.2%)	6,504 (31.3%)	-	711 (2.5%)
• Epinephrine	574 (2.01%)	-	1,933 (9.9%)	1,206 (4.3%)
• Fentanyl	5,296 (19.2%)	7,151 (34.5%)	15,340 (78.4%)	8,260 (29.3%)
• Furosemide	5,927 (21.5%)	5,320 (25.6%)	5,024 (25.7%)	8,839 (31.3%)
• Midazolam	2,851 (10.4%)	4,249 (20.5%)	4,537 (23.2%)	3,965 (14.0%)
• Norepinephrine	3,204 (11.6%)	7,336 (35.4%)	6,313 (32.3%)	4,839 (17.1%)
• Propofol	3,535 (12.8%)	13,819 (66.6%)	10,707 (54.7%)	9,457 (33.5%)

Values represent mean ± standard deviation unless otherwise indicated. For time series variables with multiple hourly recordings per ICU stay, summary statistics were computed at the stay level: the mean value across all available time points within a stay was first calculated, and Table 1 presents the distribution of these per-stay means across the cohort. This approach avoids pseudo replication arising from repeated measurements and is standard for cohort characterisation of longitudinal clinical data. Categorical variables are presented as n (%). ¹No patient identifier nor Unit Type is provided in the HiRID database; each stay was therefore considered as a unique patient. ²Values are presented as a percentage of the source database ICU stays. ³Number of ICU stays where a drug was administered at least once. *Demographic variables (age, sex, height, weight) are aggregated at the unique patient level; all other variables are aggregated at the ICU stay level. ICU: Intensive Care Unit; LoS: Length of Stay; SD: Standard deviation. A dash (-) represents a value not available in the source database

three out of four training sources showing a decline. Notably, the model trained on AmsterdamUMC shows the largest MAPE increase (+51.8%). AUPRC values for mortality prediction, reported alongside their no-skill baseline (equal to the in-ICU mortality prevalence in each

dataset), confirm and amplify the pattern observed for AUROC. All models substantially outperformed random prediction in both internal and external settings, with AUPRC values ranging from 2.5 to 4.5 times above the no-skill baseline. However, the degradation from internal

Table 2 Comparison of the TPC model's internal and external validation when training on subsets of the BlendedICU dataset

Task and training dataset	Internal validation	External validation	Variation between internal and external validation
Remaining Length of Stay prediction, MAPE [CI 95%]			
• AmsterdamUMC	80.3 [79.5–81.1]	121.9 [120.9–122.8]	+ 51.8%
• eICU	87.4 [86.8–88.0]	100.1 [99.5–100.8]	+ 14.6%
• HiRID	83.2 [82.3–84.1]	83.0 [82.6–83.4]	– 0.3%
• MIMIC-IV	87.2 [86.3–88.1]	109.5 [108.6–110.3]	+ 25.5%
Mortality prediction, AUROC [CI 95%]			
• AmsterdamUMC	0.774 [0.770–0.778]	0.808 [0.805–0.810]	+ 4.4%
• eICU	0.788 [0.783–0.793]	0.768 [0.766–0.771]	– 2.5%
• HiRID	0.851 [0.847–0.856]	0.777 [0.775–0.779]	– 8.7%
• MIMIC-IV	0.897 [0.895–0.900]	0.773 [0.770–0.776]	– 13.8%
Mortality prediction, AUPRC [CI 95%]			
• AmsterdamUMC	0.304 [0.296–0.312] (Baseline: 0.113)	0.294 [0.289–0.299] (Baseline: 0.091)	– 3.3%
• eICU	0.355 [0.344–0.365] (Baseline: 0.078)	0.269 [0.265–0.273] (Baseline: 0.107)	– 24.2%
• HiRID	0.441 [0.427–0.455] (Baseline: 0.101)	0.331 [0.327–0.336] (Baseline: 0.115)	– 24.9%
• MIMIC-IV	0.456 [0.448–0.464] (Baseline: 0.108)	0.270 [0.265–0.274] (Baseline: 0.104)	– 40.8%

Internal validation: Training and evaluation were done on separate samples from a single subset; External validation: Training was done on a sample from a single subset and evaluation was done on a sample from all other subsets

Higher AUROC and higher AUPRC indicate better mortality discrimination. Baseline AUPRC corresponds to a no-skill classifier and equals the positive class prevalence (in-ICU mortality rate) in each dataset. Lower MAPE indicates better Remaining Length of Stay prediction. 95% CI for AUROC were computed using the DeLong method; 95% CIs for MAPE and AUPRC were estimated by bootstrap resampling (10,000 iterations, percentile method)

AUROC : Area Under the Receiver Operating Characteristic curve; AUPRC : Area Under the Precision-Recall Curve; MAPE : Mean Absolute Percentage Error; CI : confidence interval

to external validation was more pronounced for AUPRC than for AUROC, with drops ranging from –3.3% (AmsterdamUMC) to –40.8% (MIMIC-IV), reflecting the greater sensitivity of AUPRC to distributional shift in the presence of class imbalance. The model trained on MIMIC-IV showed the steepest AUPRC decline despite maintaining a relatively high absolute value, suggesting

that strong internal discrimination does not guarantee robust external precision-recall performance.

The results of the internal validation were used as the baseline in the composite metric *Mcomposite*.

Comparison of the training schemes

Building on this understanding of data requirements, we compared four distinct training regimes to evaluate their impact on model specialization (MIMIC-IV and AmsterdamUMC) and generalization (unseen eICU and HiRID subsets). Specifically, we assessed: [1] single-source training on AmsterdamUMC ($N = 3,574$) [2], single-source training on MIMIC-IV ($N = 10,915$) [3], data pooling ($N = 14,489$), and [4] transfer learning ($N = 14,489$). Results are shown in Table 3.

For specialization on AmsterdamUMC, both data pooling and transfer learning substantially outperformed single-source training, with *Mcomposite* scores of 1.07 and 1.08 respectively, compared to 1.41 for training on AmsterdamUMC alone - representing improvements of approximately 24%. The two strategies, data pooling and transfer learning, performed virtually identically on mortality discrimination, with overlapping *Mcomposite* confidence intervals (1.059–1.081 vs. 1.066–1.091) and nearly equivalent AUROC values (0.75 vs. 0.76). However, data pooling demonstrated a meaningful advantage over transfer learning for RLoS prediction specifically, with a MAPE of 84.49 compared to 89.24 for transfer learning - a 5.3% improvement with non-overlapping confidence intervals (83.49–85.51 vs. 87.99–90.44). This suggests that, while both strategies yield comparable mortality discrimination at the target site, data pooling may better preserve RLoS accuracy, arguably the more operationally sensitive of the two outputs.

For generalization to unseen databases (eICU and HiRID), data pooling achieved the best performance (*Mcomposite* = 1.15), improving by 4.2% and 8.0% compared to the AmsterdamUMC and MIMIC-IV single-source models respectively. Notably, data pooling therefore performed well on both specialization and generalization simultaneously - a key finding suggesting that the trade-off between the two properties can be substantially mitigated by multi-source training.

However, data pooling incurred a performance cost for specialization on MIMIC-IV: its *Mcomposite* (0.97) was 9.0% higher than that of the MIMIC-IV single-source model (0.89), reflecting the expected trade-off when incorporating a smaller external dataset alongside a well-powered source.

Finally, regardless of the training scheme, internal validation consistently yielded overoptimistic estimates of model capability compared to external evaluation.

Table 3 Model performance on specialization and generalization in four separate training methods

Evaluation set and metric [CI 95%]	Training on AmsterdamUMC N = 3,574	Training on MIMIC-IV N = 10,915	Training by Transfer Learning ¹ N = 14,489	Training by Data Pooling ² N = 14,489
Specialization on AmsterdamUMC				
• Mortality, AUROC	0.70 [0.699–0.70]	0.72 [0.718–0.727]	0.76 [0.759–0.768]	0.75 [0.750–0.758]
• RLoS, MAPE	97.03 [95.781–98.305]	106.73 [105.321–108.139]	89.24 [87.991–90.444]	84.49 [83.492–85.514]
• <i>Mcomposite</i>	1.41 [1.398–1.423]	1.28 [1.265–1.291]	1.08 [1.066–1.091]	1.07 [1.059–1.081]
Specialization on MIMIC-IV				
• Mortality, AUROC	0.80 [0.798–0.805]	0.89 [0.892–0.897]	-	0.87 [0.872–0.878]
• RLoS, MAPE	95.92 [95.011–96.875]	86.35 [85.480–87.182]	-	87.33 [86.537–88.260]
• <i>Mcomposite</i>	1.28 [1.263–1.293]	0.89 [0.885–0.902]	-	0.97 [0.958–0.979]
Generalization				
• Mortality, AUROC	0.78 [0.776–0.784]	0.80 [0.793, 0.800]	-	0.82 [0.817, 0.823]
• RLoS, MAPE	117.08 [115.8362–118.339]	117.41 [116.275–118.576]	-	111.07 [110.042–112.174]
• <i>Mcomposite</i>	1.20 [1.189–1.212]	1.25 [1.239–1.265]	-	1.15 [1.139–1.159]

AUROC: Area Under the Receiver Operating Characteristic curve; MAPE: Mean Absolute Percentage Error; RLoS: Remaining Length of Stay; CI: confidence interval

Higher AUROC indicate better mortality discrimination. Lower MAPE indicates better RLoS prediction. 95% CIs for AUROC were computed using the DeLong method; 95% CIs for MAPE were estimated by bootstrap resampling (10,000 iterations, percentile method)

Mcomposite: a composite performance metric normalized to internal validation baseline. A value of 1 indicates performance equal to the internal validation baseline; values below 1 indicate improvement over baseline; values above 1 indicate degradation. 95% CI computed by bootstrap resampling (1,000 iterations). The composite metric is used for cross-configuration ranking only; AUROC and MAPE remain the primary performance metrics

Generalization was evaluated on the eICU and HiRID subsets, which were always unseen during training. The best performance for each evaluation set and metric is indicated in bold

¹The model was pre-trained on MIMIC-IV and fine-tuned on AmsterdamUMC. Because the aim of transfer learning is to specialize on the fine-tuned dataset, this model was not evaluated on other data sources

²Data Pooling utilized the same data as in Transfer Learning (MIMIC-IV and AmsterdamUMC), but in a single training phase

Impact of training sample size on model robustness

To determine the optimal data volume required for stable predictions, we evaluated the TPC architecture across training scales ranging from 1,000 to 55,000 patients. As shown in Fig. 2, models trained on fewer than 5,000 patients exhibited significant performance degradation across both mortality and RLoS tasks. While predictive accuracy improved with larger cohorts, a performance plateau was observed beyond 15,000 patients, where additional data yielded only marginal gains in AUROC and MAPE. The *Mcomposite* metric closely mirrored these trends, confirming its utility as a robust single indicator for benchmarking model performance. These findings established that a minimum threshold of 15,000 patients is necessary to achieve stable baseline performance, providing a quantitative justification for the sample sizes used in our subsequent multi-site experiments.

Discussion

In a clinical setting, data collection will inevitably vary between retrospective databases; for this reason, models developed for clinical usage must have strong generalization capabilities while remaining able to specialize on local data [5]. Internal validation is therefore insufficient to evaluate the performance of a model in clinical practice [7, 8]. Our study quantifies the loss of performance

during external validation and highlights the fragility of generalization when training a deep learning model using data from a single source.

Firstly, we demonstrate that internal validation is over-optimistic for evaluating clinical prediction models. Furthermore, the drop in performance varied considerably depending on the data source used for training. This “generalization gap” is driven by domain shift, including variability in patient phenotype, clinical practices, and data collection. Our results indicate that evaluating models solely on internal validation scores provides little information about their performance in clinical practice [18, 26]. This caveat regarding the evaluation of models intended for clinical applications is consistent with findings in other medical AI domains, where performance drops significantly when tested on out-of-distribution data [14, 27]. Recent work has further validated the importance of cross-institutional evaluation frameworks. Dong Jin Park et al. recently demonstrated the practical challenges of cross-hospital model deployment in a surgical context, underscoring the need for local re-validation [28]. [17] proposed a systematic benchmarking approach for generalization across ICU databases that complements our empirical findings [17]. Together, these studies reinforce the emerging consensus that external validation

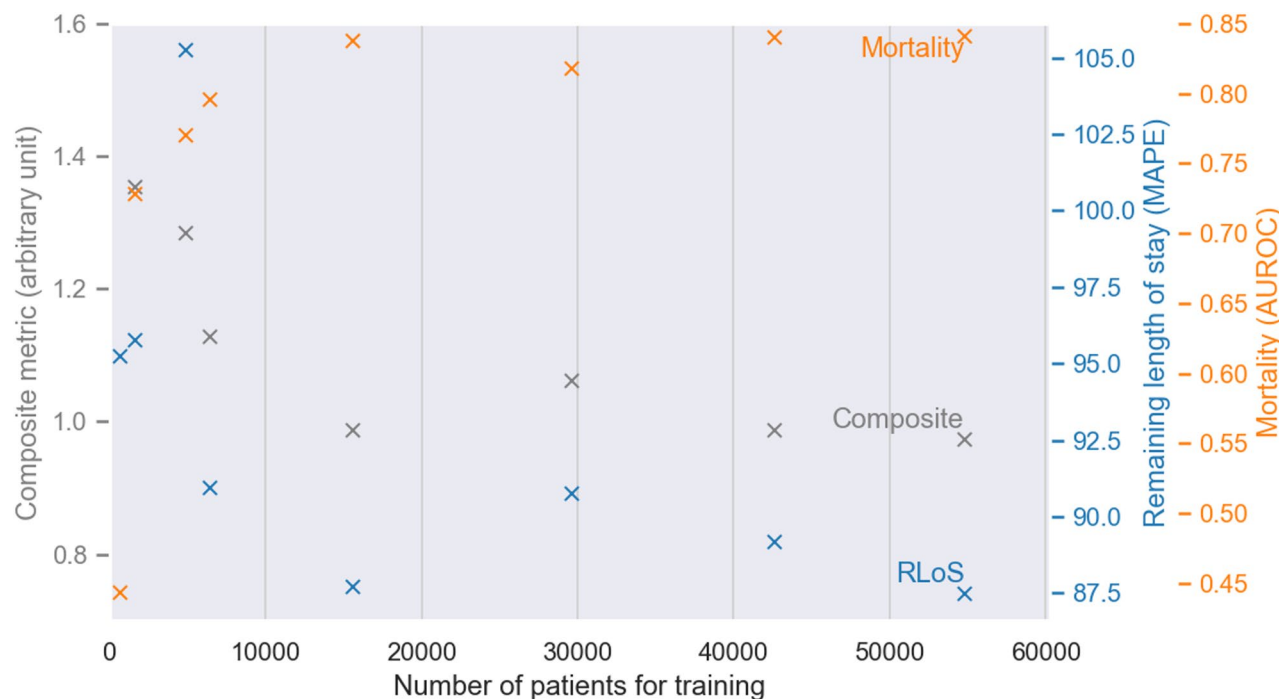


Fig. 2 Performance of the TPC model on the remaining length of stay (RLoS) and mortality prediction with increasing training size. ICU Mortality and ICU RLoS were evaluated, respectively, on the area under the receiver operating characteristic (AUROC) and on the mean absolute percentage error (MAPE); the *Mcomposite* criterion is in gray color. Data points represent discrete training sample sizes; no interpolation between points is implied or should be inferred

should be a prerequisite, not an optional step, before clinical deployment of AI models.

It is important to note the unique context of open ICU databases, which are dominated by North American cohorts such as MIMIC-IV and eICU [29]. This creates a geographic bias that limits the applicability of models in different healthcare systems, such as those in Europe. No large open-access ICU databases currently exist from Asian or African healthcare systems, which substantially limits the generalizability of models trained on predominantly North American or Western European cohorts. Beyond geographic heterogeneity, missing data patterns also differ substantially across institutions: in ICU electronic health records, the probability that a measurement is absent depends on both local data collection practices and the patient's clinical status - a Missing Not At Random (MNAR) structure that complicates model generalization across sites [22–24]. Our study therefore reinforces the debate on the value of local data. Our work also focuses on much smaller populations than the databases themselves, demonstrating that it is possible to deploy effective models based on smaller but local data sets. This suggests that high-quality, specialized local repositories may be more valuable for clinical reliability than massive, distant datasets [19, 30]. The model trained on AmsterdamUMC showed the largest degradation in RLoS prediction during external validation, with a MAPE

increase of 51.8% - the highest observed across all four training sources. This disproportionate generalization gap is noteworthy given that AmsterdamUMC has a distinctive clinical profile, including unusually high rates of certain drug exposures and specific case-mix characteristics compared to the other three databases (Table 1). It is plausible that the model learned to rely on site-specific patterns, such as local prescribing protocols or unit-type distributions, as proxies for RLoS trajectories, rather than on generalizable physiological signals. This form of shortcut learning, in which a model achieves good internal performance by exploiting features that are informative within a single institution but not transferable across sites, represents a known vulnerability of single-source training that our data cannot fully characterize but that warrants dedicated investigation before any clinical deployment of an AmsterdamUMC-trained model.

Secondly, our data pooling experiment highlighted the benefit of training on multi-site data: adding a smaller number of patients from a different data source improved generalization compared to training on a single data source. Improved generalization through data pooling was expected, as models trained on multi-site data are less prone to overfit “shortcuts” or uninformative features from a single data source [31]. This reinforces the value of harmonizing and consolidating multi-center data via the BlendedICU framework [13].

Thirdly, both data pooling and transfer learning substantially improved specialization on AmsterdamUMC compared to single-source training. While the two strategies performed virtually identically on mortality discrimination at the target site, data pooling demonstrated a meaningful advantage for RLoS prediction (MAPE 84.49 vs. 89.24, non-overlapping confidence intervals), making it the overall preferred strategy for site-specific adaptation when both outcomes are considered jointly.

While the general benefits of transfer learning are recognized, and prior studies have explored transfer learning across ICU databases for various tasks, our study is the first to systematically quantify the specialization gain from transfer learning specifically for both ICU mortality and RLoS prediction simultaneously across all four BlendedICU databases [15, 16, 32]. Transfer learning nonetheless remains a clinically valuable approach when data pooling is not feasible - for instance, when the source and target datasets cannot be shared simultaneously - as pre-training on a large retrospective database followed by fine-tuning on a small local cohort can still achieve near-equivalent specialization performance, particularly for mortality prediction.

Our findings emphasize that for AI to be safely integrated into clinical workflows, every ICU should ideally possess its own local dataset. If predictive models are shared as open-source and customizable frameworks, such local repositories would allow institutions to systematically validate external models and, where necessary, adapt them through data pooling or transfer learning depending on data availability. This decentralization of AI validation would ensure that models remain aligned with local clinical practices and patient demographics, rather than being blindly adopted from distant, potentially unrepresentative cohorts.

Regarding the model itself, this study presents the first external validation of the TPC model for ICU mortality and RLoS. In clinical practice, the two model outputs are intended to be interpreted jointly rather than independently. The hourly mortality probability provides a measure of confidence in the RLoS estimate: when mortality risk is low, the RLoS prediction carries direct operational value for discharge planning and bed allocation; when mortality risk is elevated, the RLoS estimate should be interpreted as the expected remaining stay conditional on survival, and clinical judgment must weigh this accordingly. This joint interpretation mirrors the natural framing of a bedside prognosis discussion - “what is the probability of survival, and if the patient survives, how long are they likely to remain in the ICU?” - and is more informative than either output considered in isolation. Internal performance was slightly weaker than reported by Rocheteau et al. [25]. This can be explained by our choice to prioritize real-time clinical utility. By excluding

diagnoses, which are often unavailable at the exact time of admission, and using a more realistic sample size of approximately 10,000 unique patients, our results are more representative of what is achievable in most clinical settings where massive, fully-labeled datasets are unavailable. It should also be noted that no universally accepted performance threshold exists for clinical AI tools; whether the absolute values reported here are sufficient for deployment will depend on the local clinical context, available alternatives, and cultural attitudes toward algorithmic decision support - questions that require prospective evaluation beyond the scope of this study.

Our findings are consistent with, and extend, a growing body of literature on cross-database generalization of clinical AI models. Table 4 provides a structured comparison of key studies addressing ICU clinical outcome prediction and cross-database generalization. Moor et al. examine generalization across the same four databases for sepsis prediction and include transfer learning experiments. Rockenschaub et al. systematically evaluate generalization for mortality, AKI, and sepsis across the same databases, varying the number of training centers but without transfer learning. [17] address generalization and specialization for mortality, length of stay, sepsis, and AKI, and include transfer learning experiments [15–17]. Zhao et al. demonstrate the value of a local, multi-hospital dataset in Hong Kong, achieving high predictive performance with temporal validation across 11 years - precisely the kind of local institutional validation framework our study advocates [33]. A systematic review and meta-analysis by Rockenschaub et al. confirmed that overoptimistic internal validation is pervasive in ICU machine learning research: of 572 eligible studies, only 14.7% underwent external validation, a proportion that reached only 23.9% by 2023 despite a sharp rise in publications. Among externally validated models, AUROC dropped by an average of -0.037 when applied to new hospitals, with nearly half of studies showing a clinically meaningful performance loss. Strikingly, external validation was overwhelmingly concentrated on two US databases - MIMIC and eICU were used together in 83.3% of validated studies - raising serious concerns about the geographic representativeness of current validation practices. Our study directly addresses this blind spot by conducting systematic external validation across all four of the major international ICU databases, including two European cohorts (HiRID and AmsterdamUMCdb), thereby providing evidence of generalizability that extends well beyond the North American-dominated literature [6]. Our work adds three specific contributions not addressed by these studies in combination: [1] we explicitly quantify the performance cost of external validation - the AUROC, AUPRC and MAPE gap between internal and external performance - as a primary metric

Table 4 Comparison of key studies on ICU clinical outcome prediction and cross-database generalization

Study (year)	Databases	Endpoints	External validation	Transfer learning	Temporal validation	Performance cost quantified	Composite metric	Training sample size impact assessed
Rocheteau et al. 2021 [25]	MIMIC-IV, eICU	Mortality, RLoS	No	No	No	No	No	No
Moor et al. 2023 [15]	MIMIC-IV, eICU, HiRID, Amsterdam	Sepsis	Yes	Yes	No	Indirectly	No	No
Rockenschaub et al. 2024 [16]	MIMIC-IV, eICU, HiRID, Amsterdam	Mortality, AKI, sepsis	Yes	No	No	Indirectly	No	Partial (center)
van de Water et al. 2024 [17]	MIMIC-IV, eICU, HiRID, Amsterdam	Mortality, LOS, sepsis, AKI	Yes	Yes	No	Indirectly	No	No
Park et al. 2025 [28]	2 Korean hospitals (postoperative)	Mortality, prolonged LOS (binary > 30d)	Yes, same country only	No	No	Indirectly	No	No
Zhao et al. 2026 [33]	15 HK hospitals (local)	Mortality, LOS	Yes, same system only	No	Yes	No	No	No
Allyn et al. <i>This study</i> , 2025	MIMIC-IV, eICU, HiRID, Amsterdam (BlendedICU-normalized)	Mortality, RLoS	Yes <i>All 4 DBs</i>	Yes <i>+ data pooling</i>	No	Yes, Primary metric <i>AUROC, AUPRC & MAPE gap quantified</i>	Yes <i>Generalization-Specialization trade-off</i>	Yes, <i>Minimum threshold identified (N ≥ 15,000)</i>

AKI = acute kidney injury; AUROC: Area Under the Receiver Operating Characteristic curve; AUPRC: Area Under the Precision-Recall Curve; LOS = length of stay; MAPE: mean absolute percentage error; RLoS = remaining length of stay. Yes = feature present; No = not addressed

across all four BlendedICU databases simultaneously; [2] we provide the first systematic assessment of the impact of training sample size on model robustness in this context, identifying a minimum threshold of approximately 15,000 patients for stable performance - a practically actionable finding for institutions considering local model development; and [3] we introduce a composite metric that operationalizes the generalization–specialization trade-off in a unified framework. Local datasets should ideally span several years to enable temporal validation - that is, evaluating model performance on data collected after the training period - which is a prerequisite for detecting performance drift before clinical deployment and which our current design does not address.

This study has some limitations. Firstly, it included a limited data scope: for example, model performance may be improved by adding drug dosages instead of binary exposure. Secondly, while our harmonization effort through BlendedICU was extensive, differences in sampling frequencies and missing data management across the four databases could still introduce small approximations. However, this study did not focus on feature engineering, but rather on the impact of training population diversity on model robustness. Thirdly, our demonstration concerns one particular deep learning architecture (TPC); while the authors believe this

specialization-generalization trade-off is not model-specific but rather data-specific; further research is necessary to confirm this intuition.

Additionally, the two-stage prediction architecture introduces a residual conditional bias. The model is trained on all patients regardless of outcome - including those who ultimately die - preserving the full range of illness trajectories. However, the RLoS loss is computed only for surviving patients during training: for patients who die before ICU discharge, only the mortality classification loss contributes to parameter updates, and the RLoS prediction is not evaluated. As a result, the model has never been optimized to predict RLoS for patients who deteriorate and die, meaning that the RLoS estimate for high-mortality-risk patients should be interpreted with caution, as a conditional estimate assuming survival, rather than an unconditional prediction. Alpha-weighting in the binary cross-entropy loss corrects class imbalance in the mortality classification task but does not resolve this conditional bias on RLoS prediction. Furthermore, missing data in this study are best characterized as Missing Not At Random (MNAR), and no imputation method can fully eliminate the resulting bias; our approach mitigates but does not remove it [22–24]. The available open-access ICU databases are predominantly from North American and Western European healthcare systems; no large open-access ICU databases currently

exist from Asian or African institutions, which limits the geographic generalizability of our findings. In addition, the outcome studied was restricted to in-ICU mortality; in-hospital mortality, which is a broader and arguably more patient-important endpoint, could not be studied consistently across all four databases as HiRID and eICU-CRD do not provide reliable follow-up information after ICU discharge. Our experimental design relies on a random patient-level split, which does not account for temporal trends such as changes in patient demographics, disease prevalence, or prescribing practices over time - a phenomenon known as concept drift. Temporal validation, in which the model is evaluated on data collected strictly after the training period, represents a more realistic proxy for real-world deployment performance and should be incorporated in future validation studies before any clinical implementation is considered. There is currently no universally accepted statistical framework for comparing regression-based performance metrics such as MAPE across models; bootstrapped confidence intervals were used to convey uncertainty, but formal significance testing between configurations was not possible. Finally, there is no universally accepted threshold for “good enough” predictive performance in clinical AI; the clinical utility of our models in specific deployment contexts, and their acceptability across different cultural and healthcare settings, requires prospective evaluation beyond the scope of this study.

Conclusion

Training and validating AI models on a single care site is insufficient. Our study provides large-scale empirical evidence that internal validation consistently overestimates real-world performance, and that this gap can be substantially reduced through multi-source data integration. Data pooling emerged as the superior strategy overall, achieving the best generalization to unseen databases while also matching or outperforming transfer learning for site-specific specialization, particularly for RLoS prediction. Transfer learning remains a viable alternative when joint training is not operationally feasible. These findings support a “*Data-Centric AI*” approach: the greatest performance gains arose not from model complexity but from thoughtful harmonization of international cohorts. We advocate for the democratization of local ICU datasets and the release of open, customizable models, empowering clinicians to independently validate and calibrate AI tools to their specific environment, ensuring both the safety and the portability of machine learning in critical care.

Abbreviations

AI	Artificial Intelligence
AUROC	Area Under the Receiver Operating Characteristic curve
AUPRC	Area Under the Precision-Recall Curve

APACHE	Acute Physiology and Chronic Health Evaluation
ICU	Intensive Care Unit
LSTM	Long Short-Term Memory
MAPE	Mean absolute percentage error
TPC	Temporal Pointwise Convolution
RLoS	Remaining Length of stay
SOFA	Sequential Organ Failure Assessment

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13054-026-06034-5>.

Additional file 1

Additional file 2

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Author contributions

All authors contributed substantially to the conception and design of the study, the acquisition of data, or the analysis and interpretation of the data. They also drafted (MO and JA) or provided (CF, TB, RC and NA) critical revision of the article. All authors provided final approval of the version submitted for publication.

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Data availability

The data used for this study are publicly available from their respective repositories: MIMIC-IV and eICU-CRD via PhysioNet, HiRID via PhysioNet, and AmsterdamUMCdb via the Amsterdam UMC Data Catalogue.

Availability of data and materials

The data used for this study are publicly available from their respective repositories: MIMIC-IV and eICU-CRD via PhysioNet, HiRID via PhysioNet, and AmsterdamUMCdb via the Amsterdam UMC Data Catalogue.

Declarations

Ethics approval and consent to participate

The institutional review boards of the Beth Israel Deaconess Medical Center granted a waiver of informed consent and approved the sharing of the research resource.

Consent for publication

Not applicable. The individual data presented in this article were anonymized and not identifiable.

Competing interests

The authors declare no competing interests.

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